UNIT-1: PROBABILITY AND RANDOM PROCESS

Q.1: Two numbers are chosen at random from among the numbers 1 to 10 without replacement. Find the probability that the second number chosen is 5.

Q.2 Consider the binary communication channel shown in Figure below. The channel input symbol X may assume the state 0 or the state 1, and, similarly, the channel output symbol Y may assume either the state 0 or the state 1. Because of the channel noise, an input 0 may convert to an output 1 and vice versa. The channel is characterized by the channel transition probabilities p_1, q_0 , p_1 and q_1 defined by

$$p_0 = P(y_1/x_o) \text{ and } p_1 = P(y_o/x_1)$$

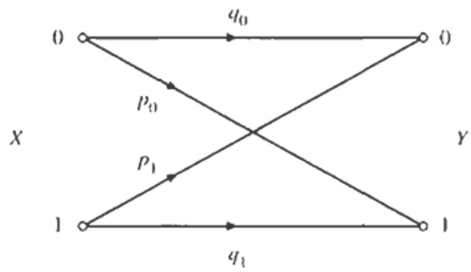
$$q_0 = P(y_o/x_o) \text{ and } q_1 = P(y_1/x_1)$$

where x_o and x_1 denote the events (X = 0) and (X = I), respectively, and y_o and y_1 denote the events (Y = 0) and (Y = 1), respectively. Note that $p_o + q_o = 1 = p_1 + q_1$. Let $P(x_o) = 0.5$, $p_o = 0.1$, and $p_1 = 0.2$.

(a) Find $P(y_o)$ and $P(y_1)$.

(b) If a 0 was observed at the output, what is the probability that a 0 was the input state?

(c) If a 1 was observed at the output, what is the probability that a 1 was the input state?



Q.3: Let X be the random variable having CDF given below

$$F_X(x) = P(X \le x) = \begin{cases} 0 & x < 1 \\ \frac{1}{2} & 1 \le x < 2 \\ \frac{3}{4} & 2 \le x < 3 \\ 1 & x \ge 3 \end{cases}$$

(a) Sketch the CDF of $F_X(x)$ of X

(b) Find (i) $P(X \le 1)$ (ii) $P(1 < X \le 2)$ (iii) $P(X > 1)$ (iv) $P(1 \le X \le 2)$

Q.4: Suppose a discrete r.v. X has the following pmfs:

$P_X(1) = ½$, $P_X(2) = ¼$, $P_X(3) = 1/8$ and $P_X(4) = 1/8$

(a) Find and sketch the CDF $F_X(x)$ of the random variable X

(b) Find (i) $P(X \le 1)$, (ii) $P(1 < X \le 3)$, (iii) $P(1 \le X \le 3)$

Q.5: For a continuous time random variable X the probability density function is given below. Find the CDF of the continuous random variable X.

$$f_X(\mathrm{x}) = \begin{cases} \dfrac{x}{2} & 0 \le x \le 1 \\ \dfrac{1}{2} & 1 < x \le 2 \\ \dfrac{1}{2}(3-\mathrm{x}) & 2 < x \le 3 \\ 0 & otherwise \end{cases}$$

Q.6: If ten coins are thrown simultaneously. Find the probability (i) of getting exactly three heads (ii) of getting atleast three heads (iii) of getting atmost three heads

Q.7: A submarine attempts to sink an aircraft carrier. It will be successful only if two or more torpedoes hit the carrier. If the submarine fires three torpedoes and the probability of a hit is 0.4 for each torpedo, what is the probability that the carrier will be sunk?

Q.8 :A continuous random variable *X* has the PDF given below(a) Find moment generating function of X. (b) Find E[X] and var[X].

$$f_X(x) = \begin{cases} \frac{1}{3}e^{-x/3}, & x > 0 \\ 0, & otherwise \end{cases}$$

Q.9: Let X be the random variable with the following pdf.

$$f_X(\mathbf{x}) = \frac{1}{2}e^{-|x|} \quad \text{for all } \mathrm{x}$$

Find CDF of Y if $Y = X^2$

Q.10: Let X be a continuous random variable X with pdf

$$f_X(x) = \begin{cases} kx & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

Where k is a constant

(a) Determine the value of k and sketch $f_X(x)$

(b) Find and sketch the corresponding CDF $F_X(x)$

(c) Find $P(1/4 < X \leq 2)$

Q.11: Consider a function

$$f(x) = \frac{1}{\sqrt{\pi}} e^{(-x^2 + x - a)} \qquad -\infty < x < \infty$$

Find the value of a such that f(x) is a pdf of a continuous random variable X.

Q.12: All manufactured devices and machines fail to work sooner or later. Suppose that the failure rate is constant and the time to failure (in hours) is an exponential random variable X with parameter λ

(a) Measurements show that the probability that the time to failure for computer memory chips in a given class exceeds 10^4 hours is e^{-1}. Calculate the value of the parameter λ

(b)Using the value of the parameter λ determined in part (a), calculate the time x_o such that the probability that the time to failure is less than x_o is 0.05.

Q.13: Binary data are transmitted over a noisy communication channel in block of 16 binary digits. The probability that a received digit is in error as a result of channel noise is 0.01. Assume that the errors occurring in various digital positions within a block are independent.

(a) Find the mean and the variance of the number of errors per block.

(b) Find the probability that the number of errors per block is greater than or equal to 4